Assignment 3

Official/Public Speaking

Speech Script for Official/Public Presentation

Duration: ~3-4 minutes
Tone: Formal, confident, and informative

Introduction

Good [morning/afternoon/evening], everyone.

My name is **Ayush Shailendra Jagtap**, and I am here with my teammates **Harshal Dhumse, Omkar Jaykar** and **Pavankumar Kalyane**, from the Department of Computer Science at **AISSMS Institute of Information Technology, Pune**.

Today, we are presenting our mini project titled:

"Performance Evaluation of RIP and OSPF Routing Protocol"

Under the guidance of **Dr. Kishor Wagh**, our respected project mentor.

◆ Problem Statement

This research aims to evaluate and compare the performance of RIP and OSPF based on key metrics such as convergence time, throughput, and routing overhead to determine their suitability for different network scales and conditions.

Project Overview

This project focuses on the comparative performance analysis of two widely used dynamic routing protocols: RIP (Routing Information Protocol) and OSPF (Open Shortest Path First). The objective is to evaluate how each protocol performs under various network scenarios using simulation tools. Key performance parameters such as convergence time, throughput, routing overhead, and packet delivery ratio are analyzed.

By simulating different network topologies and traffic conditions, the study highlights the strengths and limitations of both protocols. The results provide insights into their suitability for different network sizes and applications, supporting better decision-making in network design and implementation.

Tools and Methods

To evaluate the performance of RIP and OSPF routing protocols, the simulation-based approach was adopted using Cisco Packet Tracer, a widely used network simulation tool developed by Cisco Systems. This tool allows the design, configuration, and testing of virtual networks with realistic behavior.

Methodology followed:

1. **Network Design:**
   Different network topologies (small, medium, and large-scale) were created using routers, switches, and end devices.
2. **Protocol Configuration:**
   RIP and OSPF were configured separately on identical network setups to maintain fairness in comparison.
3. **Traffic Simulation:**
   Data traffic was generated across the network using simulated end-device communication to observe protocol behavior under load.
4. **Performance Metrics:**
   The protocols were evaluated based on key metrics including:
   - Convergence time
   - Packet delivery ratio
   - Routing overhead
   - Bandwidth usage
   - Scalability and response time
5. **Data Collection & Analysis:**
   Observations were recorded through Packet Tracer's simulation mode and event logs. These were analyzed to compare the efficiency and performance of each protocol.

Outcome and Applications

The study successfully demonstrated that OSPF outperforms RIP in terms of faster convergence, better bandwidth utilization, and improved scalability. RIP, while simpler and easier to configure, showed limitations in larger and more dynamic networks due to its slower convergence and periodic updates. The simulation results confirmed that OSPF is more efficient and reliable for complex and high-traffic network environments.

Application:

- The findings can assist network administrators in selecting appropriate routing protocols based on network size and performance requirements.
- Useful in academic training and network design projects to understand practical differences between distance-vector and link-state routing.
- Can serve as a reference for enterprise network planning, especially in scalability and protocol efficiency evaluations.
- Supports curriculum development for networking courses involving protocol performance analysis using simulation tools.

Conclusion

In conclusion, the comparative analysis of RIP and OSPF using Cisco Packet Tracer highlights OSPF as the more efficient and scalable protocol, especially for larger networks.

We thank our mentor **Dr. Wagh** and our institute for their support and guidance.

Thank you for your attention.*We are now open to questions.*